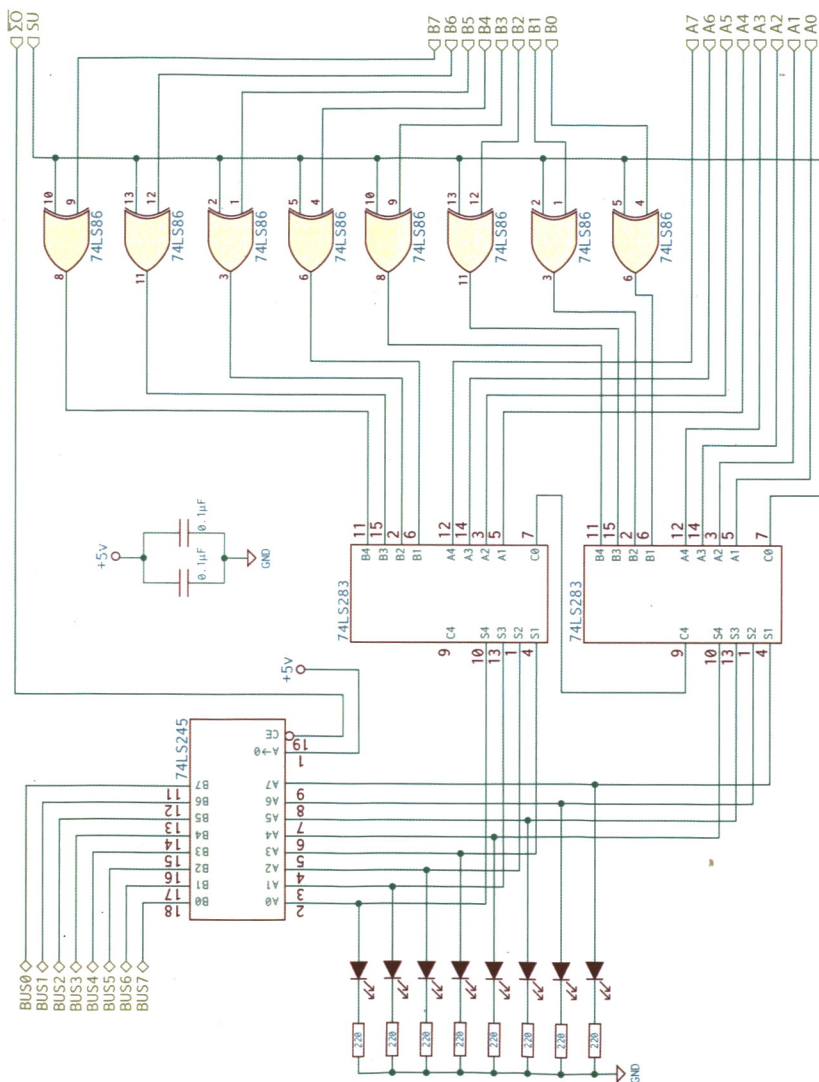


ALU schematic



If you get stuck

There's no denying it: Building a computer from scratch on breadboards is a big project. The best part is that by the time you're done, you'll have an incredible understanding of how the computer works. But at least some of that understanding will come because you had to track down some (possibly frustrating) problems. The good news is you're now part of a community of people working on this project and helping each other out. There are several good places to look for help:

- Comment threads on the YouTube videos.
- Comment threads at eater.net/8bit.
- Questions and discussions at reddit.com/r/beneater.



Caution:

You should never connect LEDs directly to a 5 volt source without a current limiting resistor. Doing so will damage the LED.

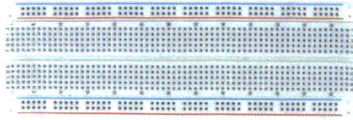
The outputs of most 74LSxx logic chips have internal resistors that let you to connect LEDs directly to the outputs without a resistor. (The 555 timer, 74189 RAM, 28C16 EEPROM, and 74HC595 *do not*.)

For the most reliable results, you may want to connect a 220Ω resistor in series with every LED anyway. I omitted many of these as a shortcut in the videos, but I've included enough 220Ω resistors in these kits if you choose (or need) to use them.

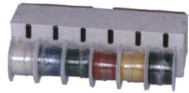
Components included in this kit

Note that some components might look slightly different than shown here or in the videos.

4 Breadboards



1 Jumper wire kit



The kit includes six 25-foot spools.

2 74LS86 quad XOR gate



6 74LS173 4-bit D register



4 74LS245 8-bit bus transceiver



2 74LS283 4-bit binary adder



40 220Ω resistors (red-red-brown) Includes 8 extra



30 Red LEDs Includes 6 extra



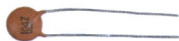
10 Yellow LEDs Includes 6 extra



5 Blue LEDs Includes 1 extra



10 0.1μF capacitors (104) Includes 5 extra



A, B, and instruction register schematic

The computer has three 8-bit registers that are essentially identical. This schematic for the A register is repeated for the B register and the instruction register. The only difference is that I used blue and yellow LEDs for the instruction register instead of red and built it on the right side of the breadboard to make things line up better in the full computer. Check out the picture on the cover of this booklet. As always, see the videos for details.

